git push

→ Accuracy, Precission, Recall, Specificity, F1 Score with 0.5

→ ROC curve with threshold at 0.7, 0.5, 0.4, 0.2

Actual Label	Predicted Score	At threshold 0.5 Predicted label	At threshold 0.2 Predicted label	At 0.4	At 0.7
1	0.85	1	1	1	1
0	0.6	1	1	1	0
1	0.7	1	1	1	1
1	0.4	0	1	1	0
0	0.55	1	1	1	0
1	0.5	1	1	1	0
0	0.65	1	1	1	0
0	0.35	0	1	0	0
1	0.6	1	1	1	0
0	0.2	0	1	0	0

Considering 0.5 ≥ → {1}

0.5 ≤ → {0}

At 0.5

	Og Pos	Og Neg
Pred Pos	4 TP	3 FP
Pred Neg	1 FN	2 TN

0.2

	OP	ON
PP	5	5
PN	0	0

0.4

	OP	ON
PP	5 TP	3 FP
PN	0 FN	2 TN

0.7

	OP	ON
PP	2 TP	0 FP
PN	3 FN	5 TN

Accuracy = $\frac{6}{10} = 0.6$

Precision = $\frac{4}{4+3} = \frac{4}{7}$

Recall / Sensitivity = $\frac{4}{4+1} = 0.8$

Specificity = $\frac{2}{2+3} = 0.4$

FPR = $1 - 0.4 = 0.6$

F1 = $\frac{1}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}} = \frac{1}{\frac{1}{0.8} + \frac{1}{0.6}} = 0.55$

~~TPR (0.5)~~

Precission = $\frac{5}{5+5} = 0.5$

Recall = $\frac{5}{5+0} = 1.0$

F1 = $\frac{1}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}} = \frac{1}{\frac{1}{0.5} + \frac{1}{1}} = \frac{1}{\frac{2+1}{2}} = \frac{2}{3}$

FPR = $1 - 0.1$

TPR = 1

pression = $\frac{5}{5+3} = \frac{5}{8}$

Recall = $\frac{5}{5+0} = 1$

TPR = 1

Pression = $\frac{2}{2}$

Recall = $\frac{2}{2+3}$

TPR = $\frac{2}{5}$

FPR = 1 − specificity

$= 1 - \frac{5}{5}$

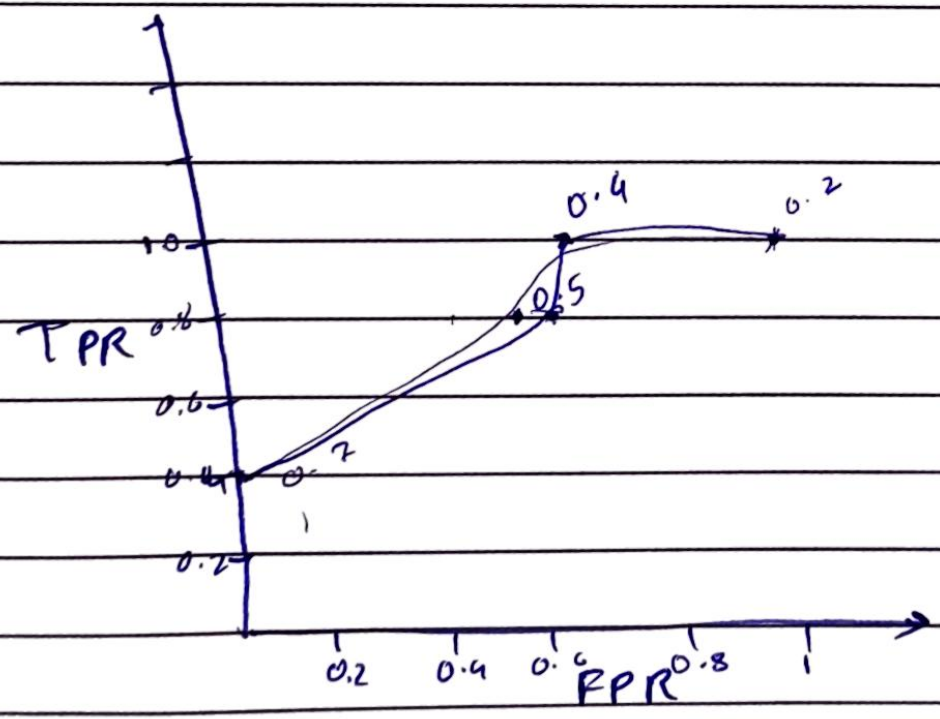
$= 0$

False PR = $1 - \frac{2}{5} = \frac{3}{5}$

F1 = $\frac{1}{\frac{1}{\text{press}} + \frac{1}{\text{real}}} = \frac{1}{\frac{2+1.6}{2}} = \frac{1}{1.8} = 0.55$

Date: / /

paperpen


TPR
FPR
1.0
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
1
0.4
0.2
0.5
0.7